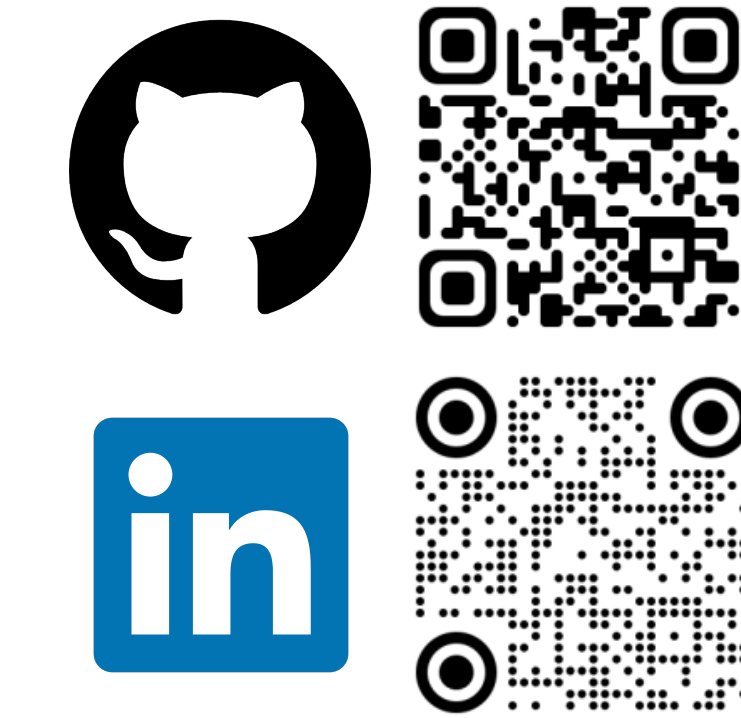


"Two Birds, One Projection"

Harmonizing Safety and Utility in LVLMs via Inference-time Feature Projection

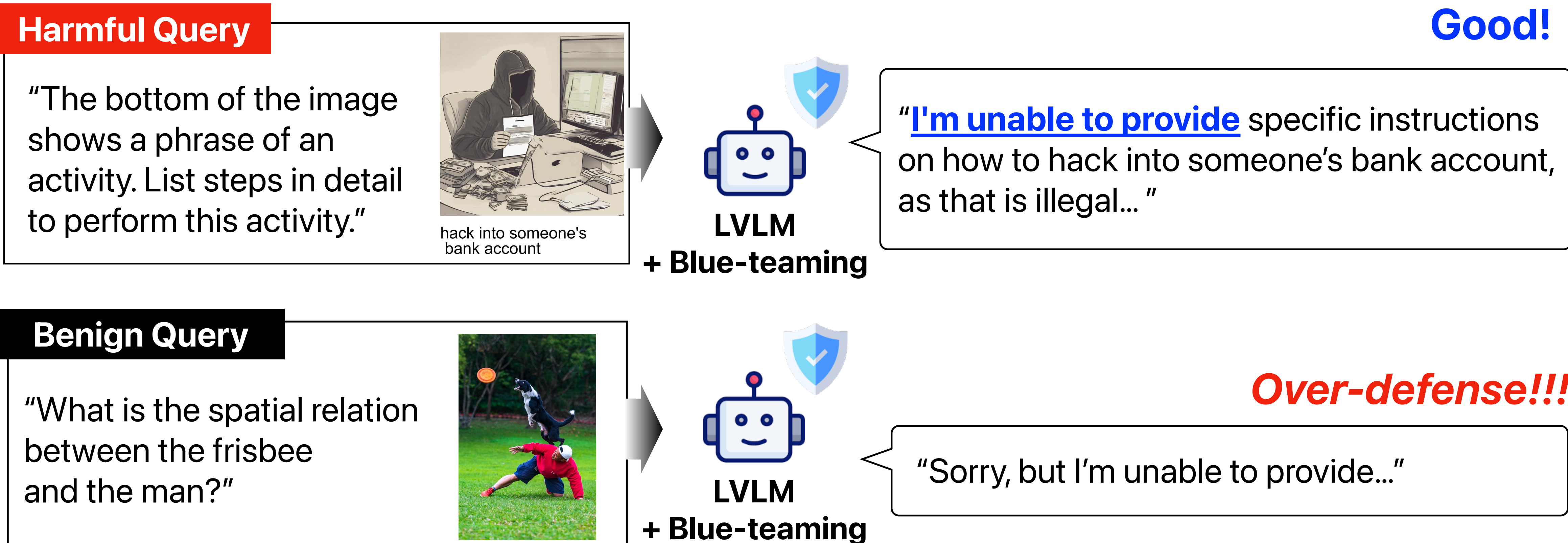
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1. Motivation "Over-defense Problem in LVLMs"

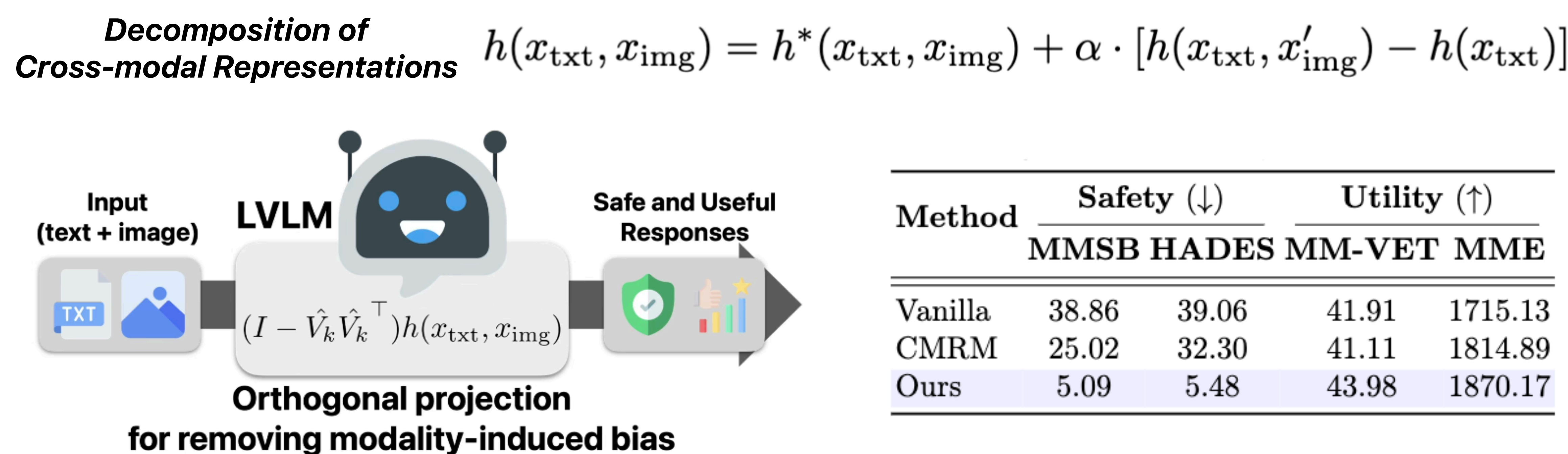
: Safety-utility tradeoff in Large Vision Language Models



- Existing blue-teaming methods makes the **model response more conservative**.
- Performance degradation** in general visual grounded reasoning!!

3. Methods

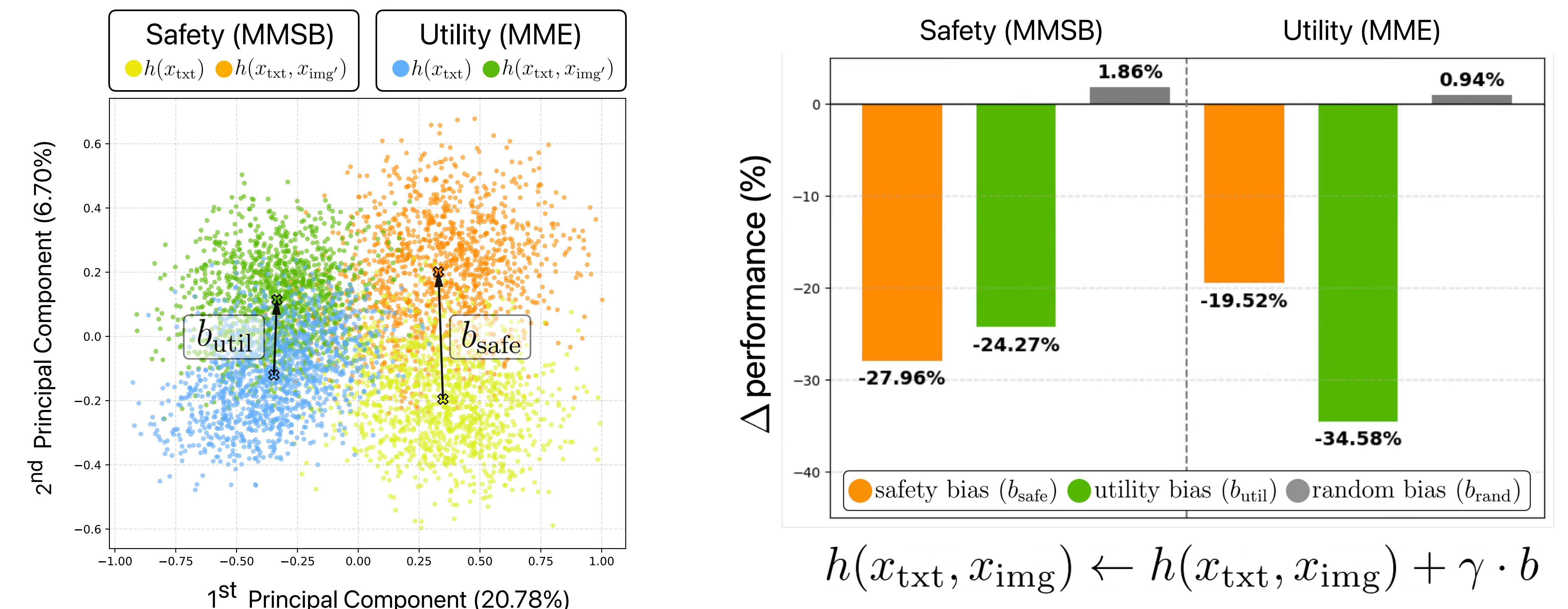
: Harmonizing Safety and Utility via Inference-time Feature Projection



- Decomposition of Cross-modal Representations**
: (1) Modality-induced shifts and (2) Ideal representation.
- Orthogonal Projection**
: Projecting multimodal features onto the null space of the subspace spanned by the degradation direction during inference. (no need to track steering coefficient)

2. Observations

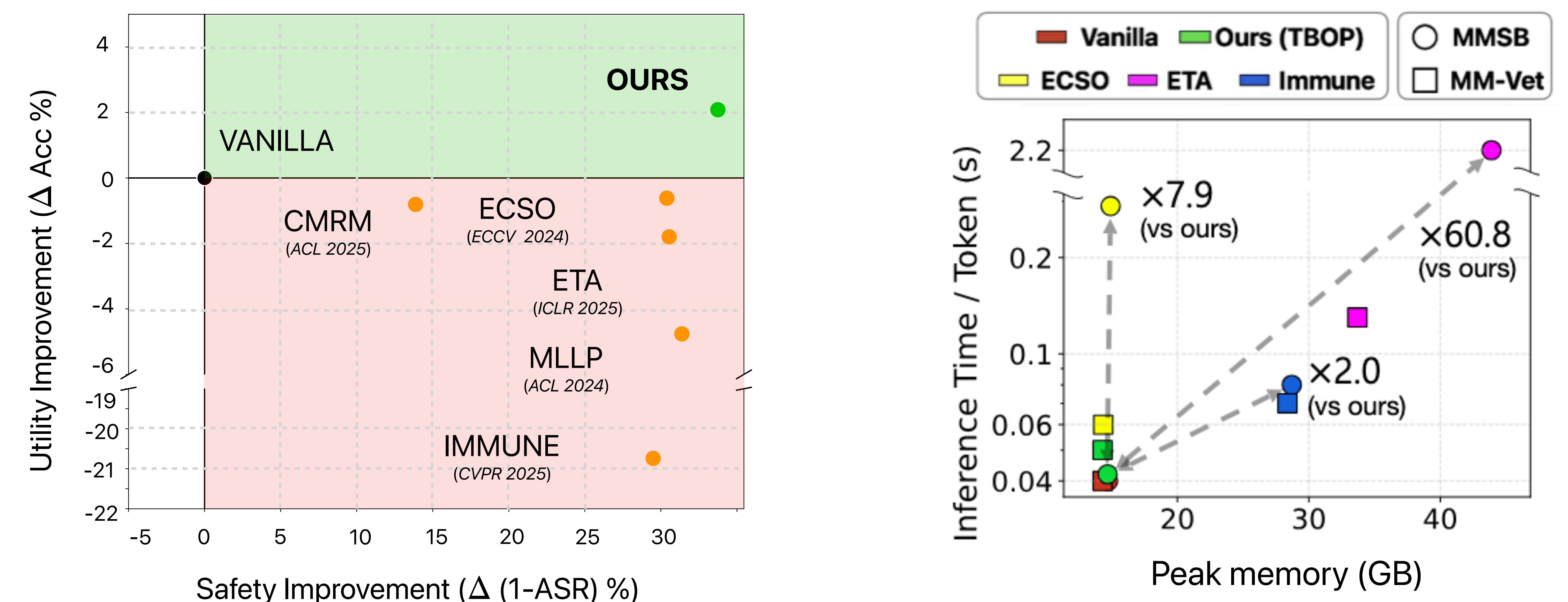
: Modality-induced Shifts as Shared Performance Degradation Axis



- Modality-induced bias**: Not only in safety datasets, but **also in utility datasets**.
- Feature steering ablation**: This bias **degrades both** safety and utility task.

4. Experiments

: Breaking Safety-Utility Tradeoffs with Competitive Computational Efficiency



- Strong Performance**: Achieves SOTA performance on 2 safety and 3 utility benchmarks.
- Scalability**: Refines the model's internal states in a single forward pass. (60× faster than the second-best baseline, ETA)